



Fig. 3: Generic design procedure for software radios

The RF stage includes output power generation, pre-amplification, and conversion of RF signals suitable for wideband analogue-to-digital and digital-to-analogue conversion.

Data converter. The data converter impacts the performance of the overall radio design. It requires very high sampling rates ($f > 2.5W$), high number of effective quantisation bits, an operating bandwidth of several GHz and a large spurious-free dynamic range. Two fundamental steps performed by data converters are sampling and quantisation.

The spurious-free dynamic range (SFDR) specification is very useful for application of the analogue-to-digital converter (ADC). It is useful when the desired signal bandwidth is smaller than the Nyquist bandwidth.

The performance of ADCs continues to improve fast. For radio receiver applications using digitisation at the RF or IF, ADCs with both high sampling rates and high performance are desired. But there is a tradeoff between these two requirements.

High-speed signal processing. This includes baseband processing, modulation, demodulation, bitstream processing, coding and decoding. It can be implemented with DSPs, application-specific integrated circuits (ASICs) and FPGAs. For example, the Motorola 68356 includes a general-purpose microcontroller with a 56000 series DSP.

DSPs use microprocessor-based architecture and support programming in HLL. ASIC implements the system circuitry in fixed silicon resulting in most optimised implementation in terms of speed and power consumption. FPGAs provide much hardware-level reconfig-

urability allowing much more flexibility than ASICs but less than DSPs. In general, the three hardware components constitute a design space that trades flexibility, processing speed

and power consumption.

Architectural characteristics

The architecture of software radio is still evolving. Till date software radios feature open system architecture, reconfigurability, flexibility, modularity, scalability, validation, authentication, replicability and cross-channel connectivity:

Open system architecture. Standard that describes the layered model of communication or distributed data processing system

Reconfigurability. Ability of radio's personality to change most commonly through reprogramming. It may include reconfiguration of the waveform from IS-95 to EDGE, changing individual algorithm or changing parameters of algorithm

Flexibility. Ability to accommodate reconfigurations without changing the system architecture

Modularity. Encapsulation of each of the various tasks that define a system into individual and separate modules, whether in software or hardware, interconnected with each other. The system can be changed through addition or replacement of individual modules without affecting the design of other modules

Scalability. Ability to add extra modules

Validation. Ensures that designed architecture is able to meet all its requirements and goals

Authentication. Ensures that the changes being made are appropriate, permitted and desired

Replicability. Ability to support the addition of new channels to the system by simply adding copies of the

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